

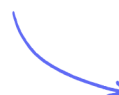
Calculator Questions

Transformations of Shapes

Translations / Reflections / Rotations / Enlargements

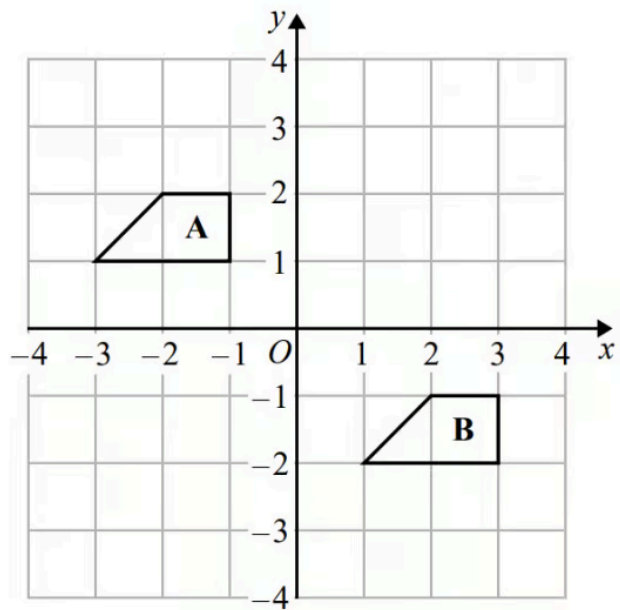
Easy (4 questions)	/9
Medium (3 questions)	/11
Hard (3 questions)	/11
Very Hard (3 questions)	/9
Total Marks	/40

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Easy Questions

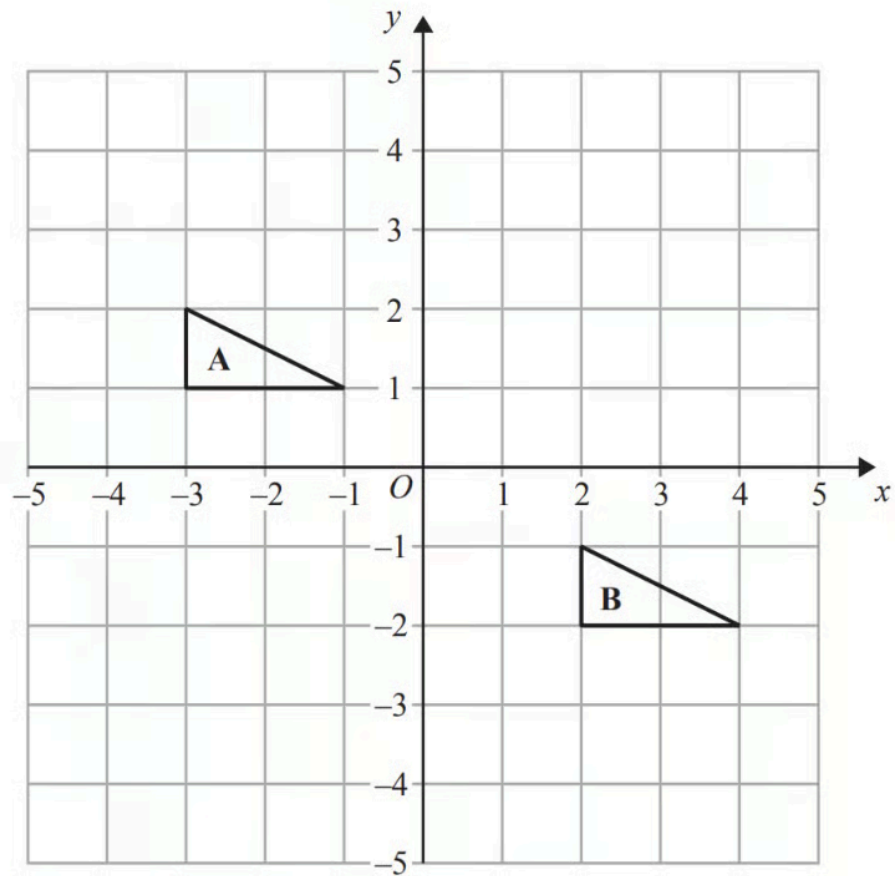
1



Describe the single transformation that maps shape **A** onto shape **B**.

(2 marks)

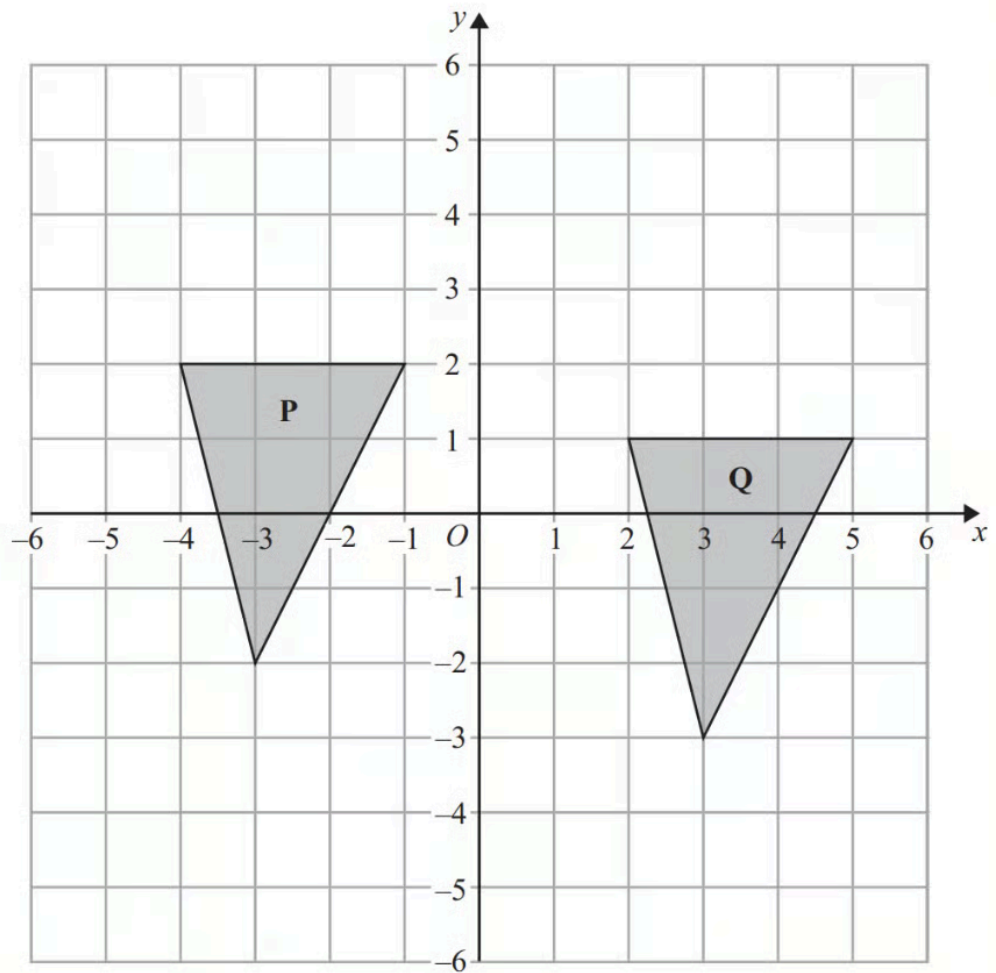
2



Describe the single transformation that maps triangle **A** onto triangle **B**.

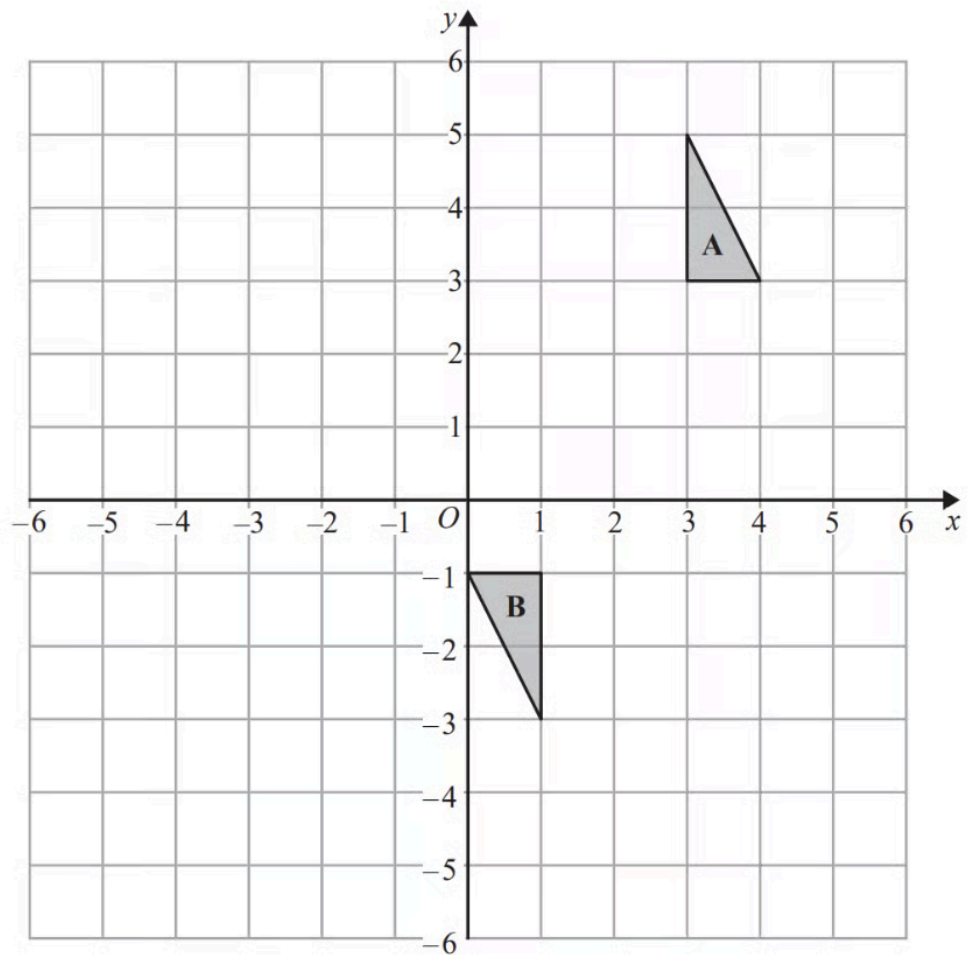
(2 marks)

3



Describe fully the single transformation that maps triangle **P** onto triangle **Q**.

(2 marks)

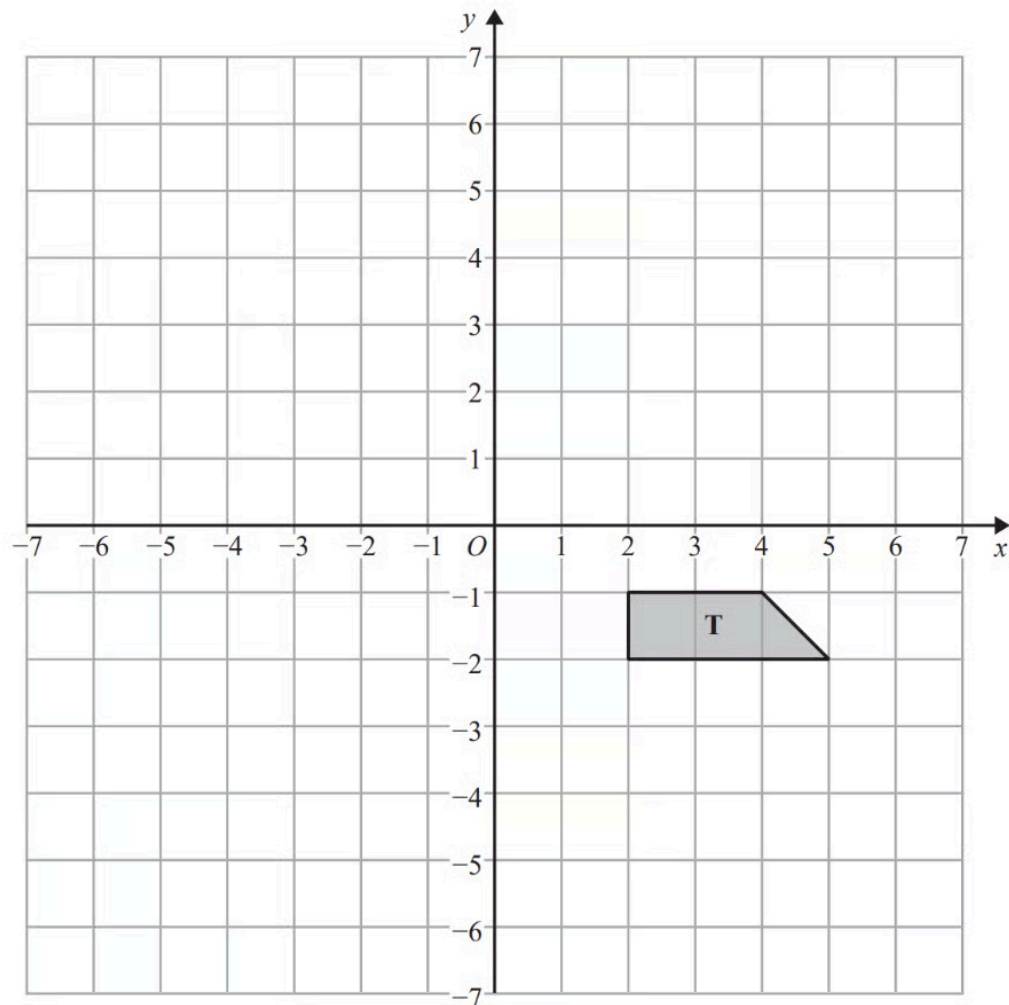


Describe fully the single transformation that maps triangle **A** onto triangle **B**.

(3 marks)

Medium Questions

1 (a)



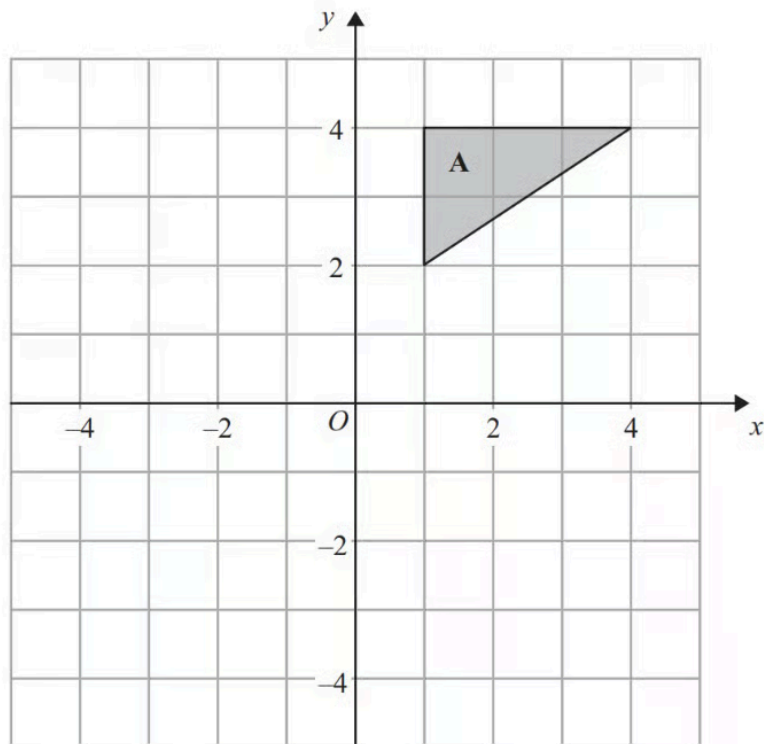
Rotate trapezium **T** 180° about the origin.
Label the new trapezium **A**.

(1 mark)

(b) Translate trapezium **T** by the vector $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$ Label the new trapezium **B**.

(1 mark)

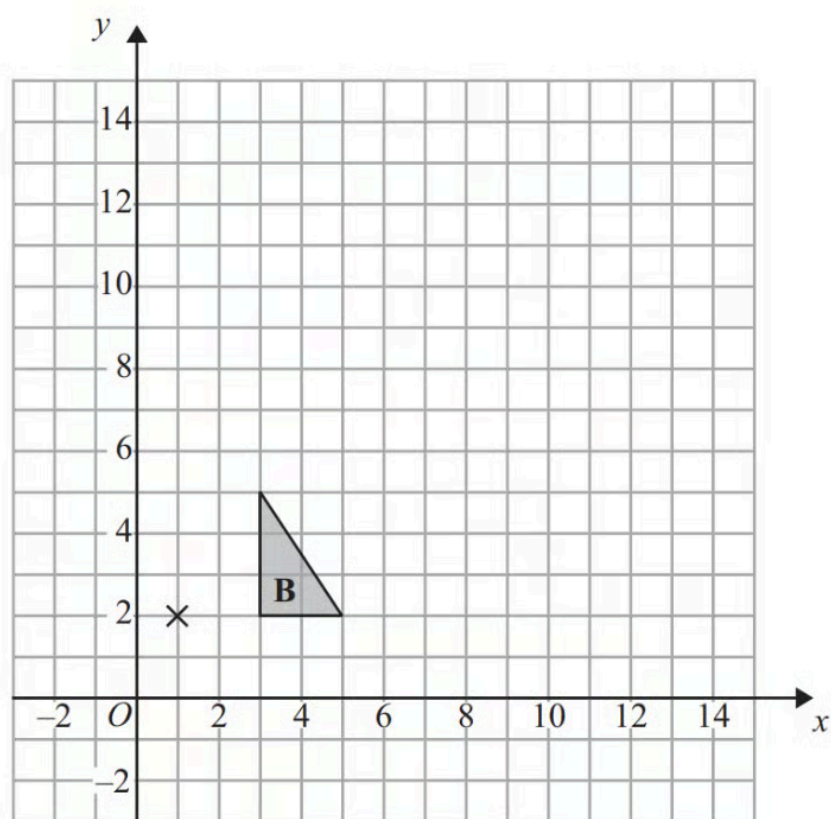
2 (a)



Rotate triangle **A** 90° clockwise, centre O .

(2 marks)

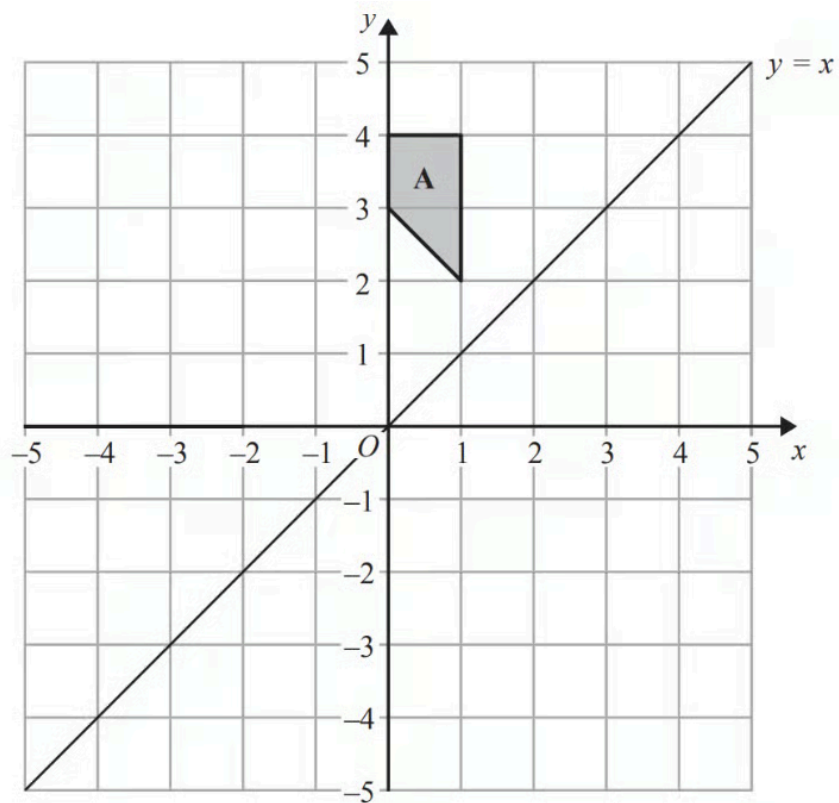
(b)



Enlarge triangle **B** by scale factor 3, centre (1, 2).

(3 marks)

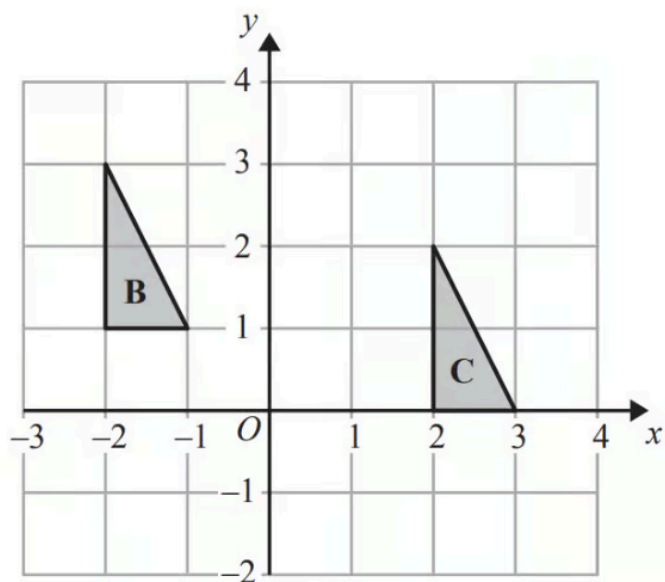
3 (a)



On the grid, reflect shape **A** in the line $y = x$.

(2 marks)

(b)

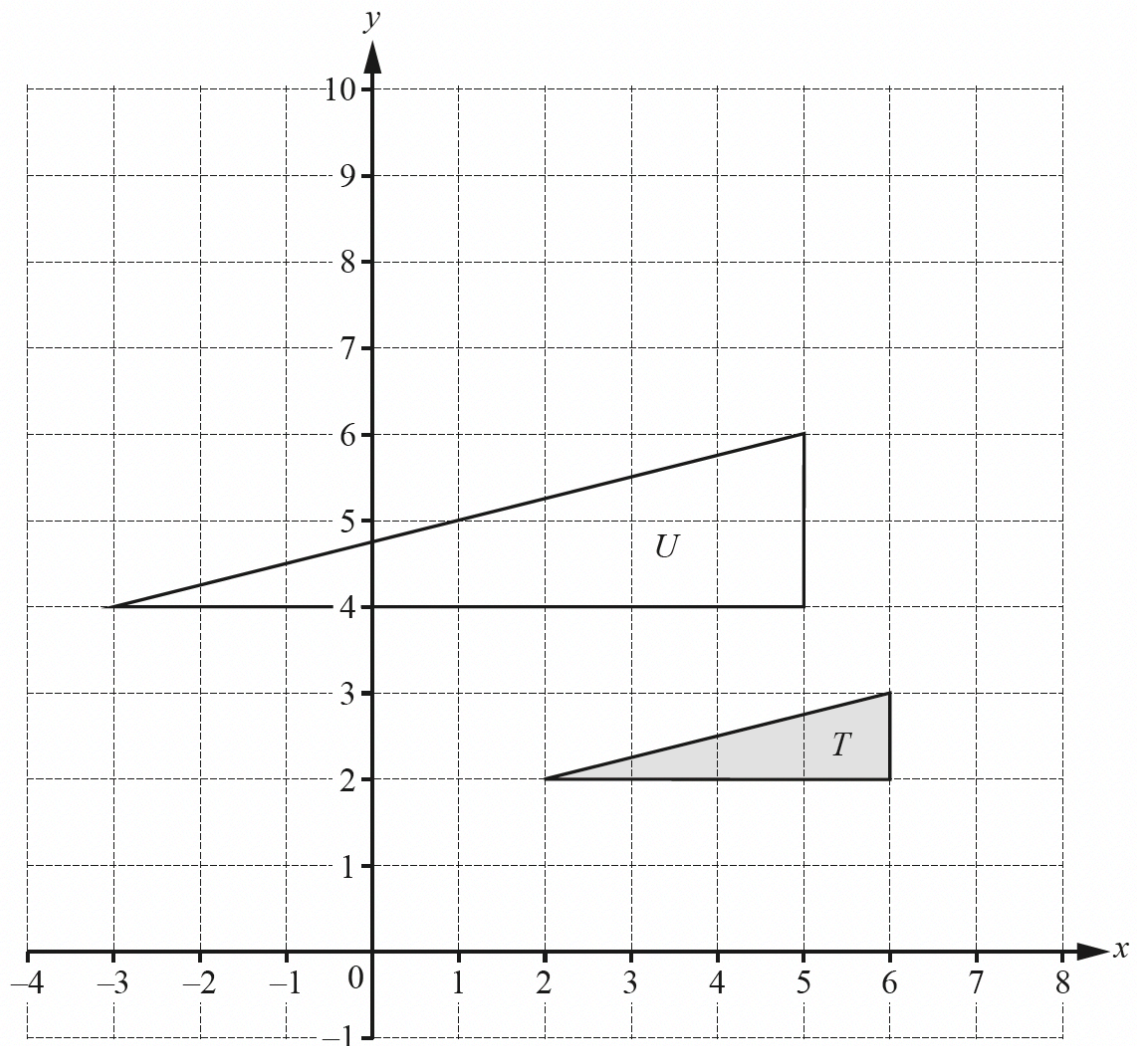


Describe fully the single transformation that maps triangle **B** onto triangle **C**.

(2 marks)

Hard Questions

1 (a)



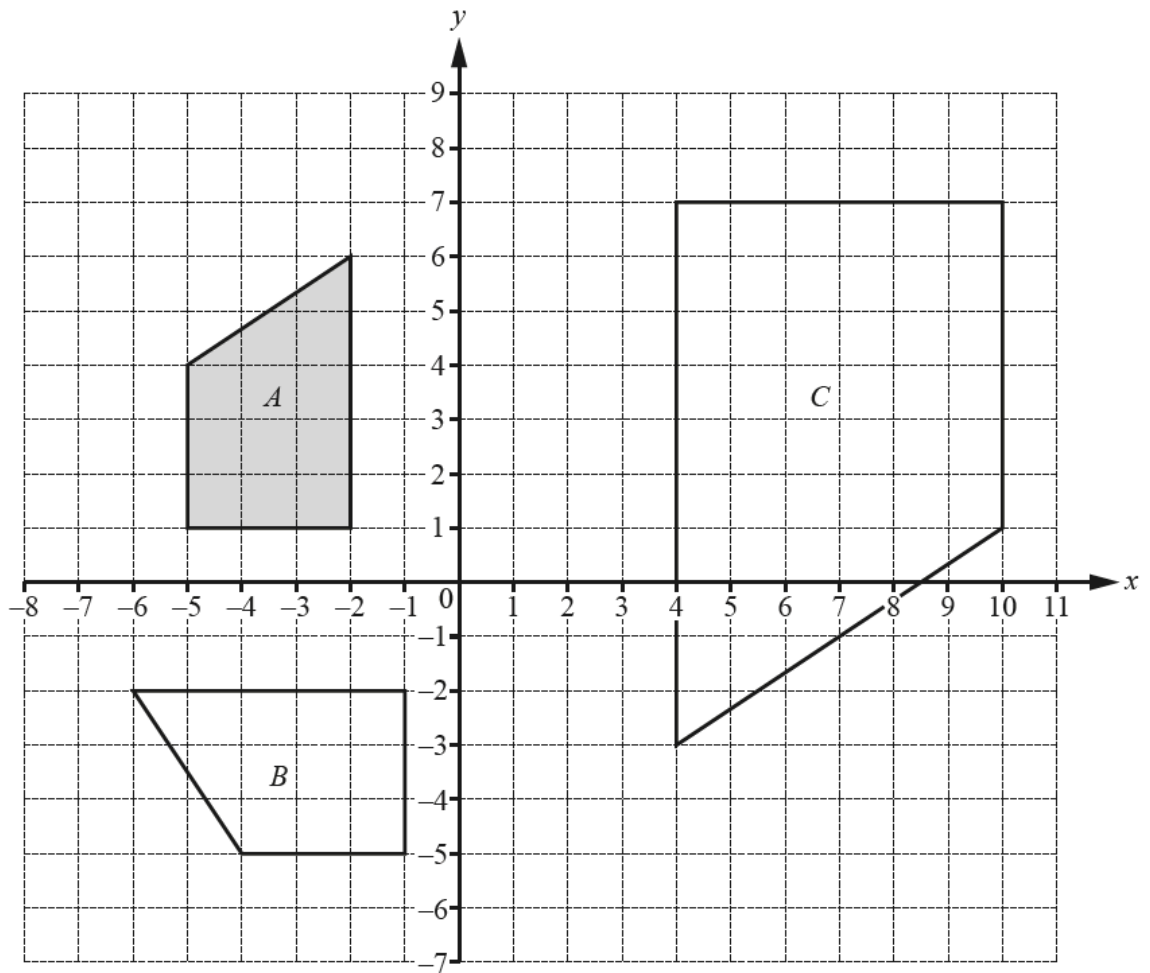
Describe fully the **single** transformation that maps triangle T onto triangle U .

(3 marks)

- (b) On the grid, draw the image of triangle T after a rotation through 90° clockwise about the point $(7, 3)$.

(3 marks)

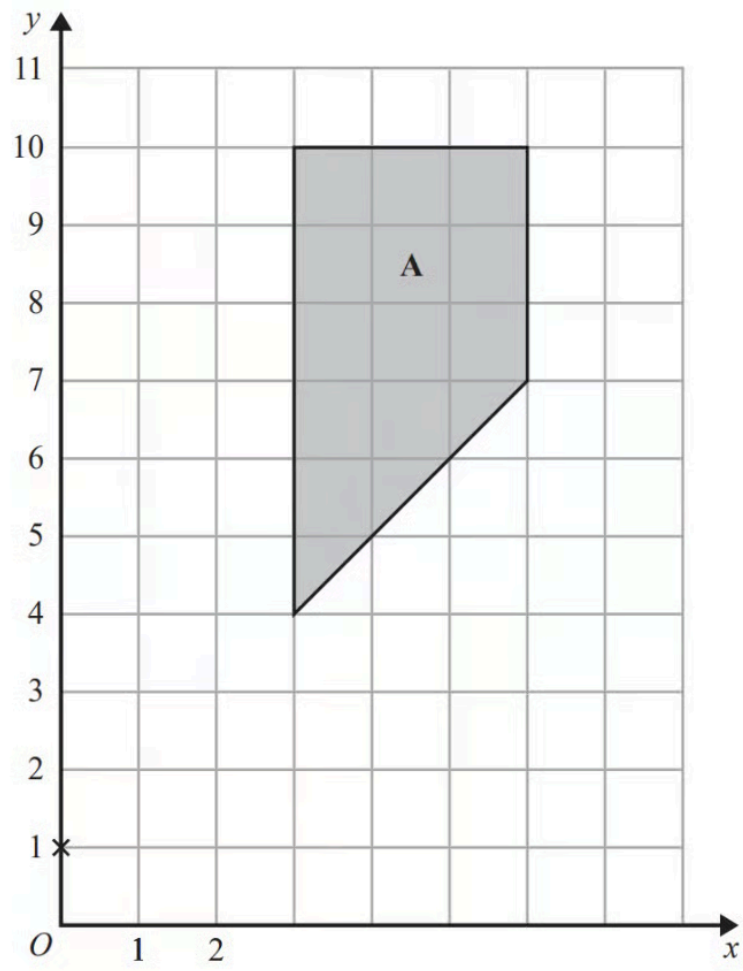
2



Describe fully the **single** transformation that maps shape A onto shape B.

(3 marks)

3

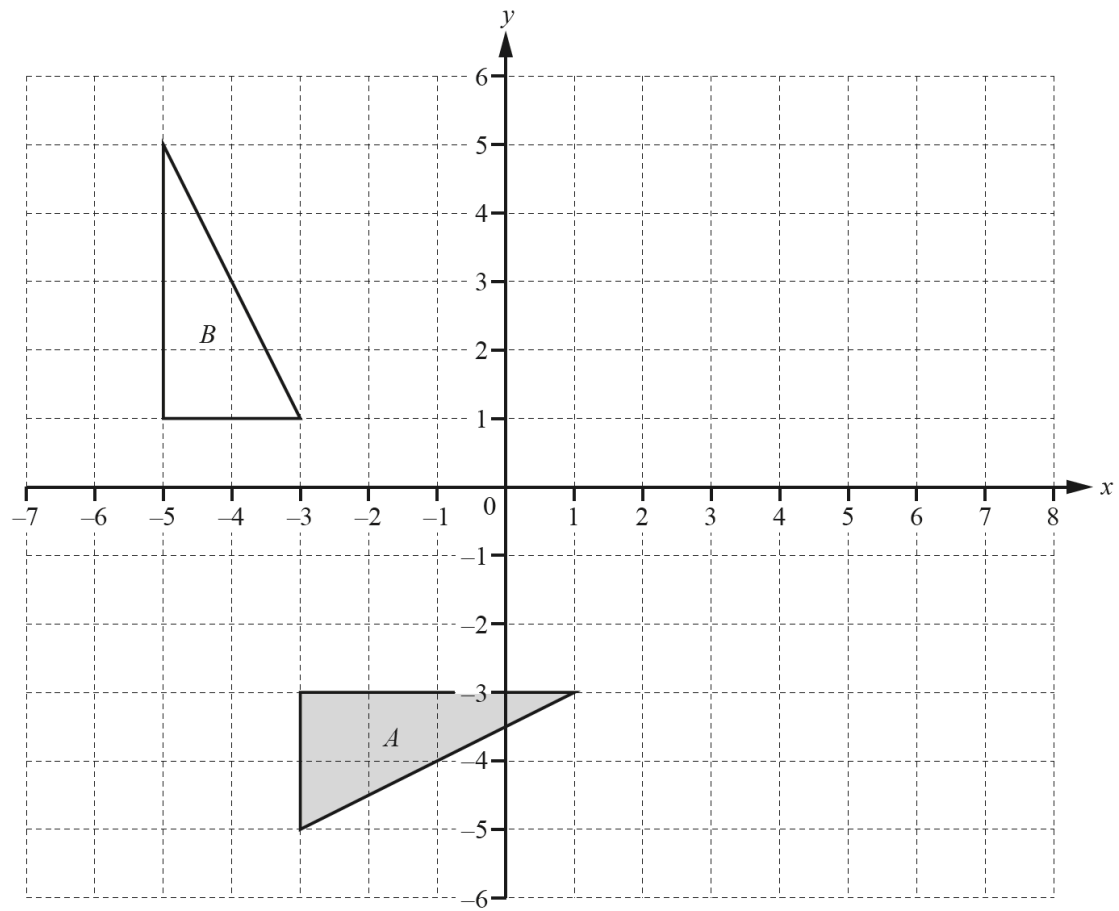


Enlarge shape **A** by scale factor $\frac{1}{3}$ centre (0, 1).

(2 marks)

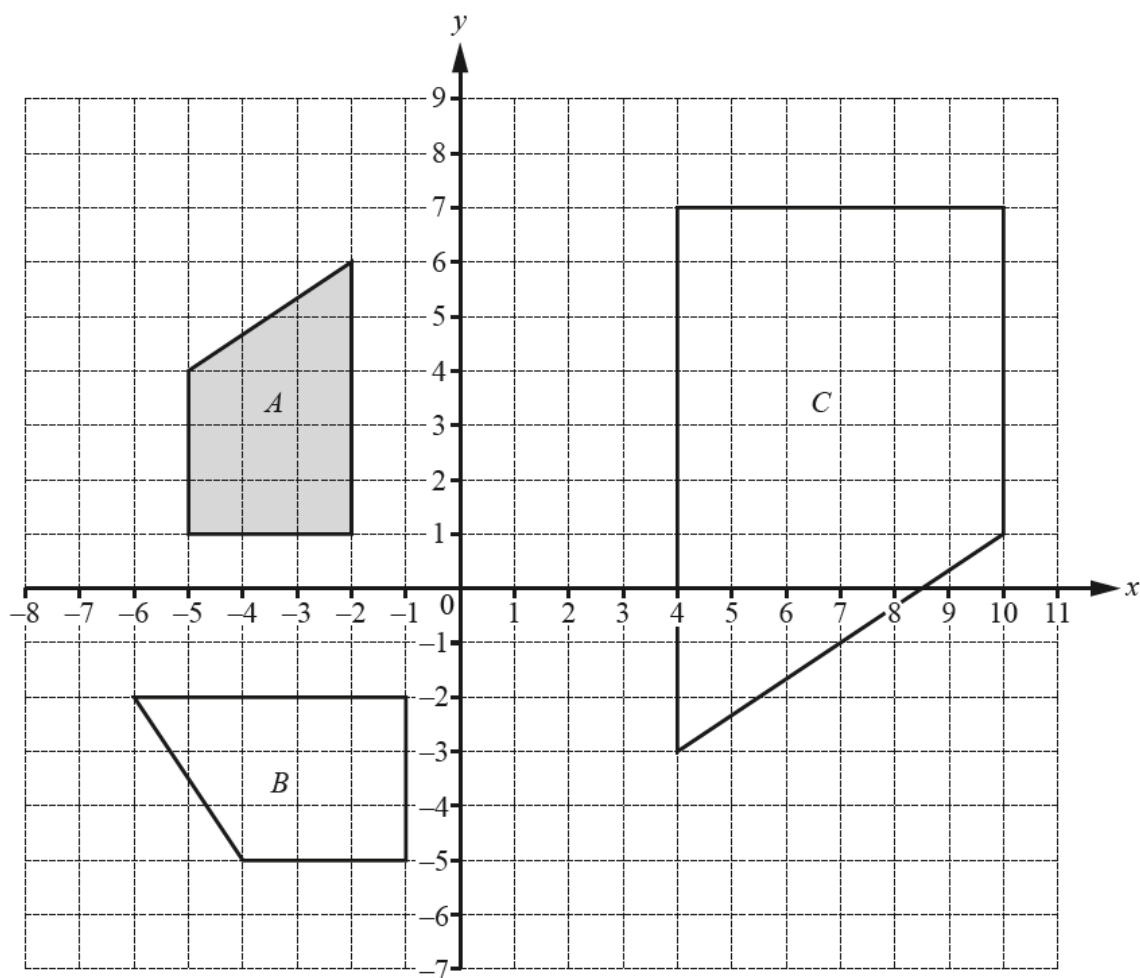
Very Hard Questions

1



Draw the image of triangle A after an enlargement by scale factor $-\frac{1}{2}$, centre $(3, 1)$.

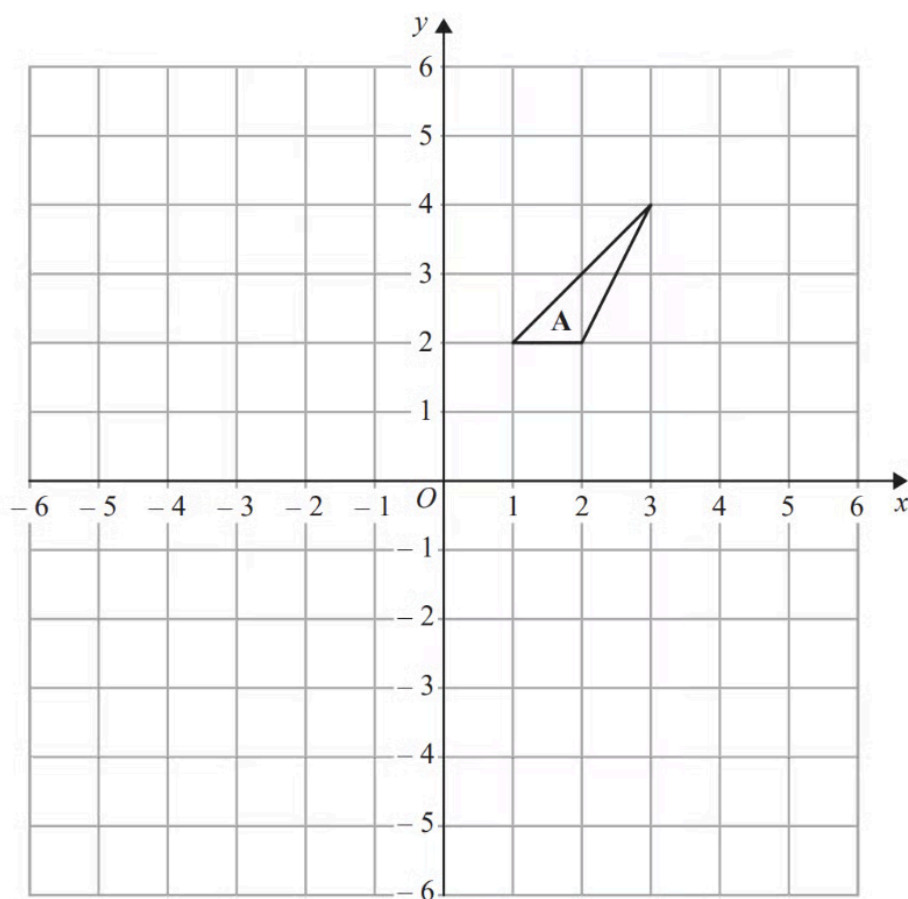
(3 marks)



Describe fully the **single** transformation that maps shape A onto shape C.

(3 marks)

3



Triangle **A** is rotated 90° clockwise about the point (0, 1) to give triangle **B**.

Triangle **B** is translated by the vector $\begin{pmatrix} -3 \\ -1 \end{pmatrix}$ to give triangle **C**.

Describe fully the single transformation that maps triangle **A** onto triangle **C**.

(3 marks)